

Embedded System Workshop (EC3.202)

Introductory Class

Instructors: Sachin Chaudhari, Aftab Hussain, Deepak G.
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INTERNATIONAL INSTITUTE OF
INFORMATION TECHNOLOGY

HYDERABAD

Background: Sachin Chaudhari (SPCRC)

- **Signal Processing and Machine Learning for Wireless Communication**
 - Internet of Things (IoT)
 - 5G and Beyond, Satellite Communications, Cognitive Radios
- **During PhD and post-doc**
 - Spectrum Sensing for Cognitive Radios
 - **Worked on EU projects Encor2** (2013-2014) and **SENDORA**
- **At IIITH**
 - Cognitive Radios for 5G and Beyond (2015-2019)
 - IoT for Smart Cities (2018-ongoing)
 - IoT-enabled Smart Cities: Pollution, Health and Governance (*DST and PRIF*)
 - CoE on IoT for Smart Cities (*India-EU Collaboration Project for Standardization*)
 - Smart City Living Lab (*MEITY and Smart City Mission*)
 - 5G Use Case Lab at IIITH (*DoT*)
 - Satellite Communication for IoT (2020-ongoing)

Background: Deepak Gangadharan (CSG)

- **Academics**

- Asst Prof., IIIT Hyderabad (2019-present)
- PostDoc Researcher: UPenn (2015-2019), University of Erlangen Nuremberg (2013-2015), Technical Univ. Of Denmark (2012-2013)
- PhD (CS), National University of Singapore (2007-2012)
- BTech (ECE), College of Engineering Trivandrum (1998-2002)

- **Industry**

Design/Senior Design Engineer, Conexant Systems Inc. (2002-2005)

- **Research Interests:**

Cyber-Physical Systems, Real-Time Scheduling, Intelligent Transportation Systems, Edge Computing

- **Research Projects**

- ITS: Traffic flow prediction and control, Vehicular Edge Computing, Digital Twin
- UAV security
- CPS: Real-Time Scheduling for Safety Critical Systems

Background: Aftab Hussain (CVEST)

- **Academics**

- Asst Prof., IIIT Hyderabad (2018-present)
- Postdoc Fellow, Harvard University (2016-2018)
- MS+PhD, KAUST, Saudi Arabia (2011-2016)
- B.Tech, IIT Roorkee (2005-2009)

- **Industry**

- Design Engineer, Analog Devices India (2010-2011)

- **Research Interests:** Flexible electronics, sensor systems, smart cities, IoT

- **Research Projects**

- Talk to me...

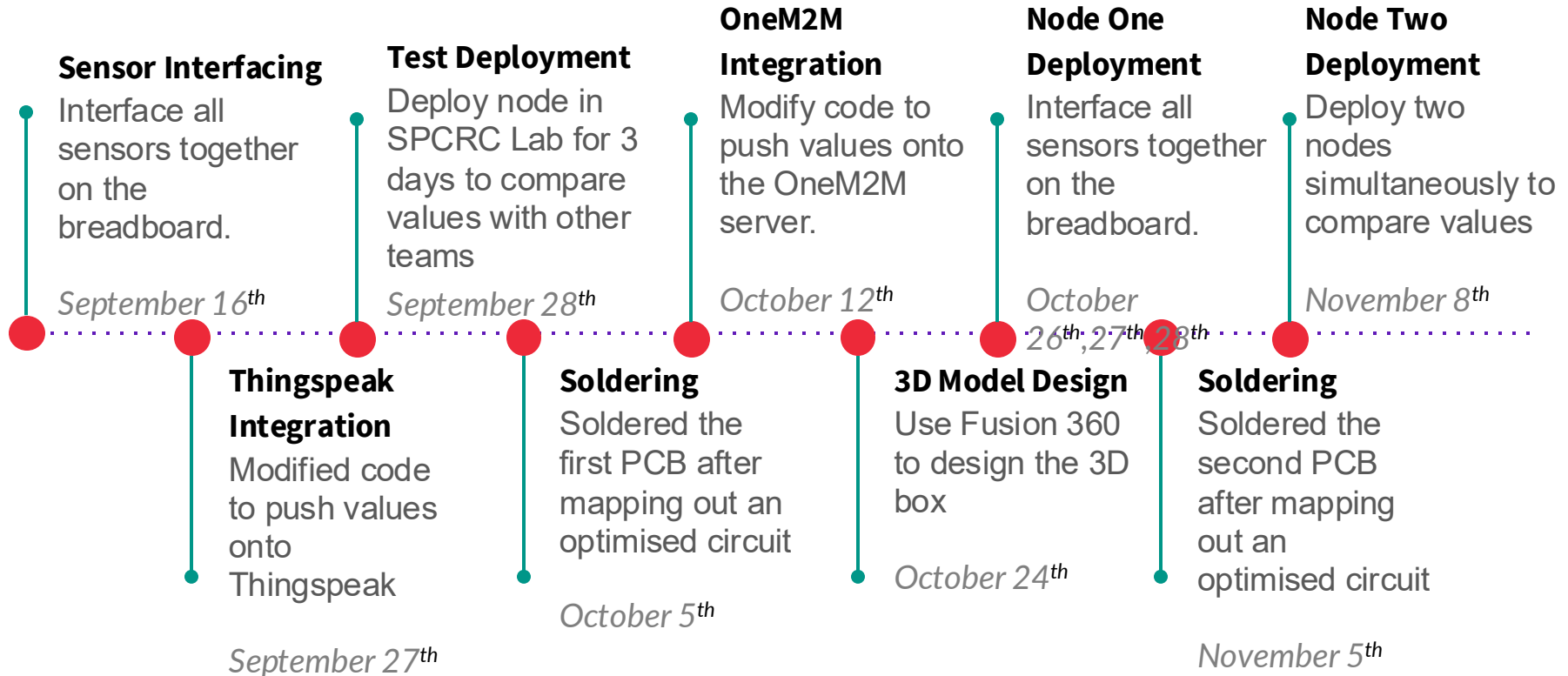
Outline

- Course Intro
 - Embedded System Workshop (Previous years)
 - Projects this time
- Course Administration
 - Syllabus
 - Resources
 - Evaluation

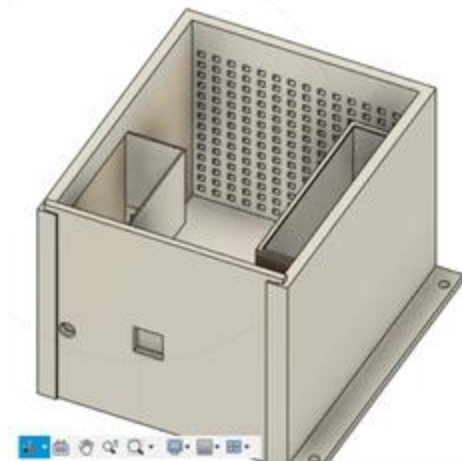
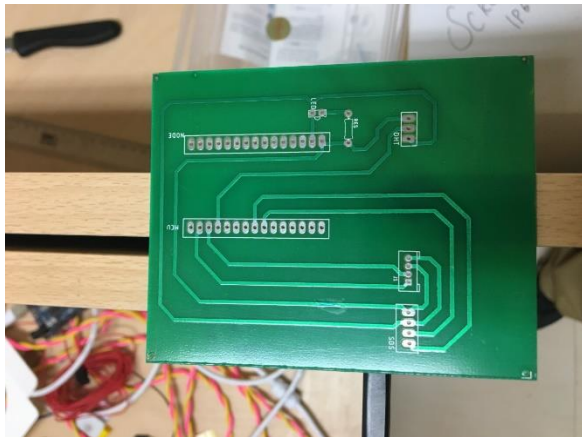
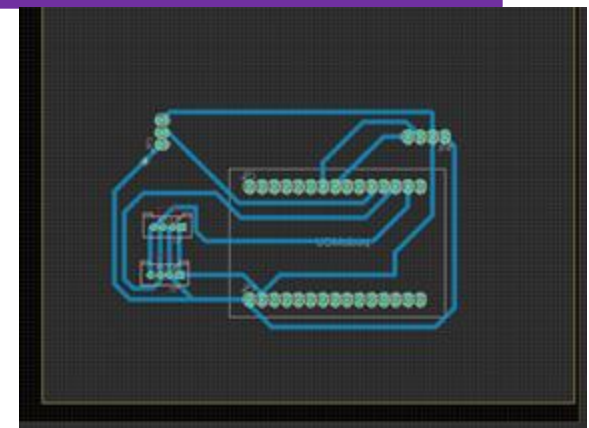
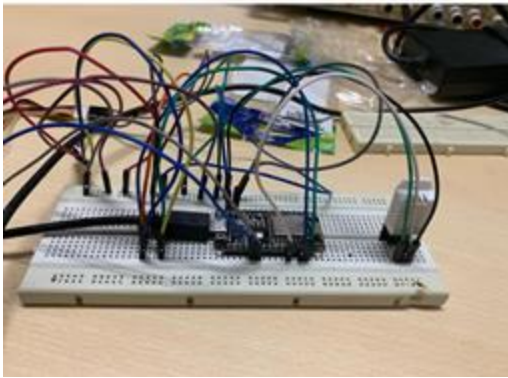
ESW: (M2019-M2021)

- First phase of Smart City Living Lab done by ESW course
 - Coordinators: Vishal Garg, Sachin Chaudhari, Aftab Hussain
 - Hardware provided by institute
 - Topics
 - Air pollution
 - Building energy efficiency
 - Water quality and Quantity
 - Smart classrooms
 - Streetlights
 - And Many more
 - Youtube: <https://www.youtube.com/watch?v=RcxKJ54GH6Q>

ESW: *Different Aspects Involved*



ESW: *Different Phases*



ESW: Some Photos from Deployment

- Indoor: Faculty Quarters – I and II



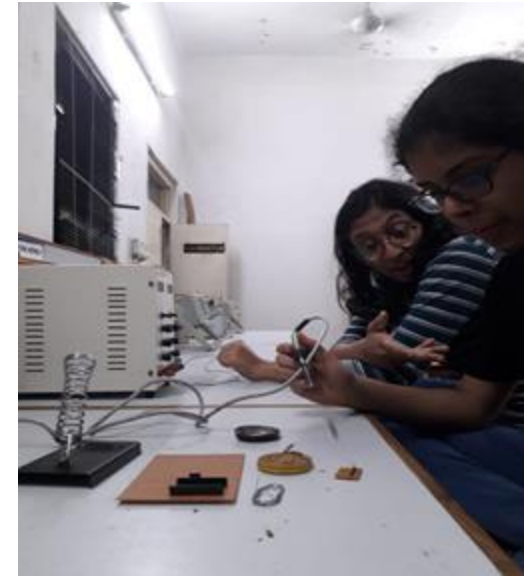
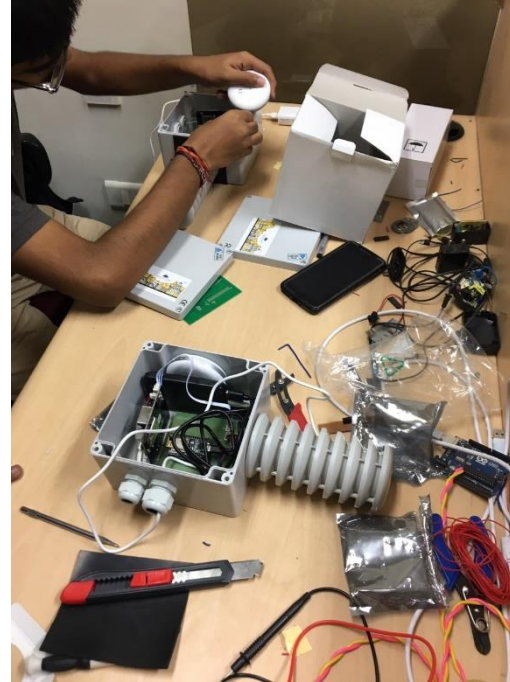
- Indoor: Mess



ESW: Outdoor Deployment

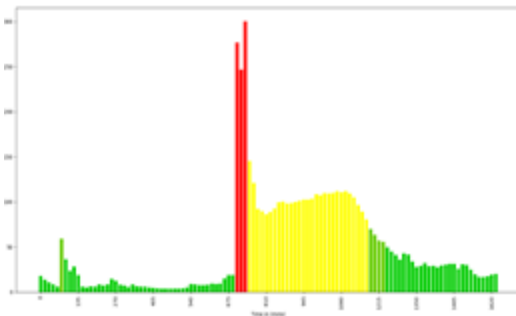
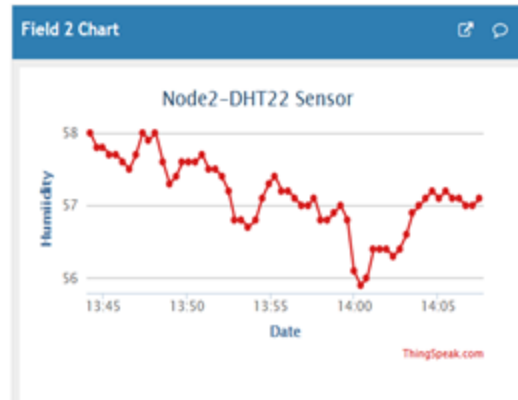
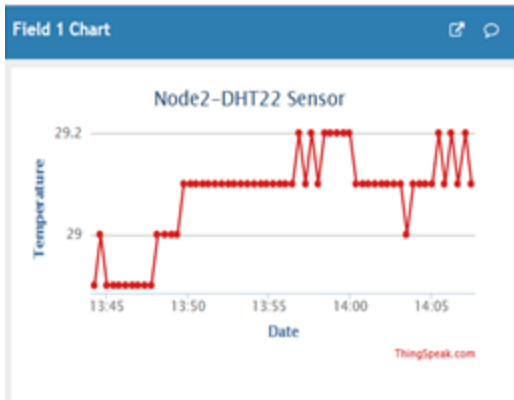
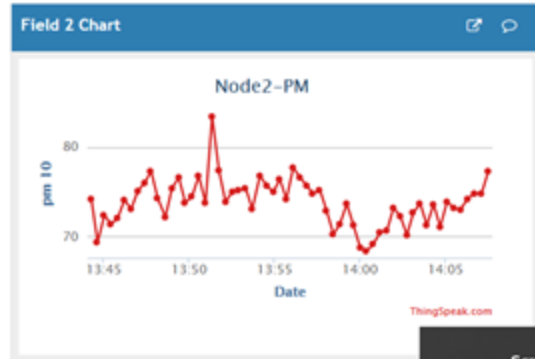
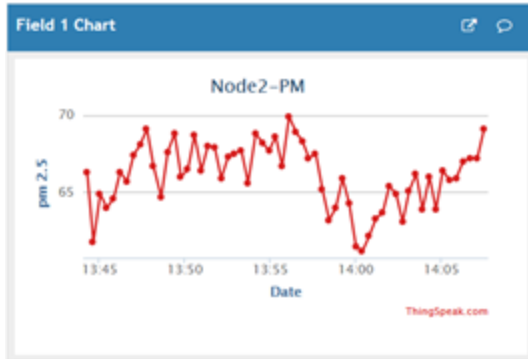


ESW: *Students at Work*



ESW: Sample Data

- ThingSpeak data -



- We have our own Twitter channel !!



oneM2M

ESW 2022-2024

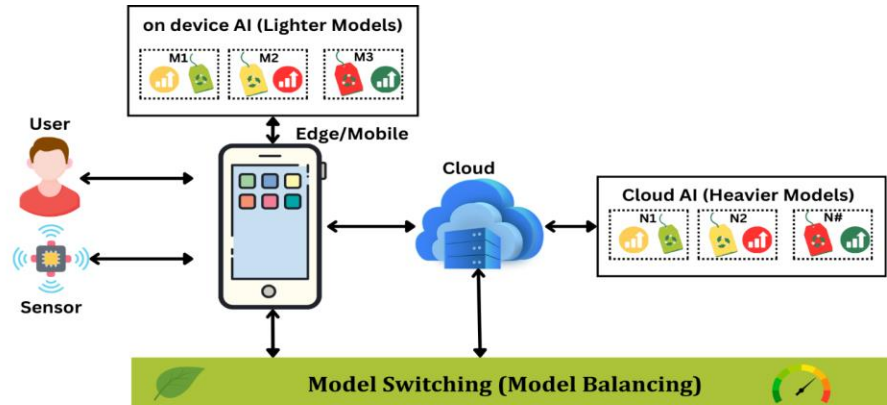
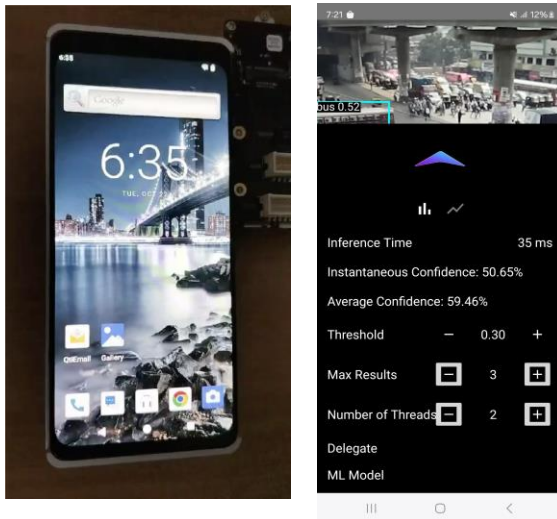
- Abhishek's comments
 - Several interesting IoT and embedded projects
- Publications
 - Inesh Dheer, Shreyas Mehta, Srikar Somanchi, Alan Nelson and Abhishek Srivastava, "Pre-Deployment Calibration Framework for Low-Cost Gas Sensors: An Adaptive Environmental Parameter Model," IEEE Sensors Letters, June 2025.
 - Rathan et. al. "Edge-Based Object Classification Using mmWave Radar on Qualcomm Innovators Development Kit" ICEdge 2025
- Introduction of Qualcomm Innovation Development Kits (QIDK) projects in Monsoon 2024 in collaboration with Qualcomm

ESW 2022-2024: QIDK projects

- Several interesting projects
 - Edge ML balancer, AQI estimation using image, pose estimation, waste segregation, Object classification using mmWave radar
 - Few publications
 - SPARQ invitation



Model Switching and Crowd Monitoring



EdgeMLBalancer: A Self-Adaptive Approach for Dynamic Model Switching on Resource-Constrained Edge Devices

Akhila Matathammal*, Kriti Gupta¹, Larissa Lavanya¹,
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Small Models, Big Tasks: An Exploratory Empirical Study on Small Language Models for Function Calling

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IIIT-Hyderabad demonstrates ML model-switching approach on smart phones for real-time traffic monitoring

Telangana

M Srinivas

27 February 2025 2:55 PM

Second year CSE students of IIIT-Hyderabad unveiled ML model



A team of 2nd year CSE students of IIIT-Hyderabad has successfully demonstrated a dynamic ML model-switching approach on smart phones for real-time traffic monitoring. (Photo: By Arrangement)

Projects are open-ended!

- Few things which can be common across **IoT projects**:
 - **Soldering, PCB, 3D Enclosure**
 - **Basic dashboard or App**
 - **Data processing / Alerts**
 - **Cloud interfacing**
 - RTC, Power management (putting device to sleep), Security
 - Data transmission and storage reduction, Managing communication outage
- Students are encouraged to test, deploy, and productize their projects
- Same is true for QIDK projects
 - Can compare with other computing platforms (QIDK 2.0, Raspberry Pi) if available
 - Try out different models
 - Can explore more possibilities in discussion with faculty and Qualcomm team

IP policy

- Faculty are giving part of their open research problems
 - They will continue doing research on this topic
 - If you do sufficient contribution, you can be part of their papers/patent
 - Significant research work needed even after basic PoC
 - Faculty's decision

Projects for ESW based on two themes:

- * IoT
- * QIDK

(Tentative themes are provided)

IoT-based projects

Project 1: Two-wheeler Event Detection

- Faculty Guide: Deepak Gangadharan
- TA: Arihant Jain
- One of the most used transportation modes in South-East Asian countries is two-wheelers or bikes. However, there is very little work that has looked into the development of algorithms and system for detecting and analyzing the riding behavior of bike riders. This is very essential for ensuring the safety and developing policies for overall transportation ecosystem. In order to understand the riding behavior, the first essential task is to detect events such as driving straight, driving over bumps, turning right and left, etc.
- Develop an embedded system to capture the vehicle dynamics using IMU. Place it in a bike and collect data with various events. Further, use the data to develop learning algorithms for classification of the events.

Project 2: Digital Stethoscope for heart and BP monitoring

- Faculty Guide: Sachin Chaudhari
- Mentor: Dr. Sandeep
- TA: Bhargava
- Traditional stethoscopes depend on the doctor's hearing and experience. This makes it hard to record or share the sounds for second opinions or further analysis. Digital stethoscopes on the market are often expensive and not suitable for primary health centers or field camps. This highlights the need for a simple, low-cost digital stethoscope that can accurately capture heart sounds and Korotkoff sounds.
- Develop a digital stethoscope for measuring heart rate and BP (Korotkoff sounds). It should also connect to a smartphone or computer. Make a simple dashboard.
- Collect data and analyze the efficacy of your device.
- Compare with existing devices

Project 3: Compost Monitoring

- Faculty Guide: Kavita Vemuri
- TA: Kunal
- Composting is a natural process that converts organic waste into valuable fertilizer. The efficiency of composting depends on maintaining ideal environmental conditions — particularly temperature and moisture. Temperature indicates microbial activity; too low means slow composting, too high can kill microbes. Moisture affects microbial metabolism and decomposition rate. pH and oxygen levels also influence compost quality but are secondary for basic monitoring.
- The goal of this project is to design and implement a system that continuously monitors key parameters of a compost pile, such as temperature, moisture, and sometimes pH and oxygen levels, to ensure optimal composting conditions. This helps in maintaining the microbial activity for efficient breakdown of organic waste into nutrient-rich compost.
- Tasks
 - Task1: Get the compost containers prepared – can use dry leaves for this project instead of food waste.
 - Task2: understand the parameters that can indicate the compost readiness
 - Task 3: prepare the list of sensors and the circuit.
 - Taks 4: install and monitor for at least 45 days. Try different waste material
 - Track the time for the compost to be prepared.

Project 4: Sleep Quality Monitoring

- Faculty Guide: Kavita Vemuri
- TA: Kunal
- Understanding sleep stages and body movements can offer valuable insights into sleep quality. Monitoring heart rate helps identify the body's physiological state, while EMG sensors track muscle activity, allowing differentiation between light and deep sleep. The gyroscope helps detect physical movements, which can be used to identify restlessness and wakefulness during the night. Together, these sensors can give a broad overview of a person's sleep quality without being overly intrusive.
- The goal of this project is to develop a sleep monitoring device that uses three key sensors—EMG (Electromyogram), heart rate, and gyroscope—to monitor sleep quality, sleep stages, and body movements during sleep. The system will analyze muscle activity, heart rate patterns, and body movement to assess sleep stages (e.g., light, deep, REM) and detect sleep disturbances such as restlessness.
- TASKS
 - Task1: Integrate & time sync the 3 sensors
 - Task2: understand the parameters that can sleep quality
 - Task 3: prepare the list of sensors and the circuit.
 - Taks 4: use the device after testing on at least 3 people, each recorded for at least 7 days.
 - Task 5: also use a quality of sleep survey that the person fills each day.

Project 5: Secondary Pollutant Estimation near Pharma Industrial Cluster (CGD)

- Faculty Guides: Kiran Chand + Sachin Chaudhari
- Mentor: Sandeep Budde
- TA: Sasidhar
- Secondary pollutants such as ozone, PAN, and aerosols are formed when primary pollutants (PM, CO₂, VOCs, NO_x, Ammonia) from industrial emissions during late nights undergo photochemical reactions in the presence of sunlight. Their concentration can vary with height due to changes in solar radiation and atmospheric dispersion.
- The project aims to calculate and compare the formation of secondary pollutants at different heights near windows of buildings close to a pharmaceutical industrial cluster using telescopic sensors.
- Measure pollutant concentrations at the given heights using the sensor. Analyze variations in pollutant levels with height and correlate with sunlight intensity.

Project 6: Estimating Vertical Pollution Dispersion (CGD)

- Project: Faculty Guides: Kiran Chand + Sachin Chaudhari
- Mentor: Sandeep Budde
- TA: Sasidhar

- Conventional air quality monitoring systems usually measure PM concentrations at the ground level, providing little information about the vertical dispersion of these particles in the atmosphere.
- To address the need for vertical PM distribution data, we deploy PM sensor nodes vertically at different heights. This innovative design allows for continuous monitoring and recording of PM concentrations at various heights, providing detailed insights into the dispersion patterns of particulate matter from the ground up.

Project 7: Internet of Medical Things

- Faculty mentor: Priyesh Shukla (CVEST and CSG)
- In collaboration with AIG (Dr Sai Tarun Vasala)
- TA: Bhargava
- AI-enabled HW-SW co-designed system for improved patient experience and remote monitoring
 - Collect EHR (electronic health record data), patient vitals from sensors (oximeter, etc.), audio inputs, motion/gestures through camera... (use of available datasets in the beginning, and later with real patient data)
 - Agentic inference and service based on the received data
 - Reasoning-based + Multimodal AI-based models for Edge deployment

Project 8: Wi-SUN Environment monitoring

- Faculty: Aftab Hussain
- TA: Advaita
- Use a simple environmental monitoring sensor – DHT22 and interface with the existing Wi-SUN network on campus
 - If done, check signal strength at various points
 - If done, check difference between FSK and OFDM

Project 9: LoRa Localization

- Faculty: Aftab Hussain
- TA: Advaita
- Use RFM95 LoRa boards to form a peer-to-peer network
- Use this to perform the path-loss experiment
 - If done, use 4 RFM95 boards to perform a triangulation experiment using LoRa signal strengths

Project 10: Edge Traffic Agent with Negotiation Protocol Platform

- Faculty mentor: Suresh Purini (CSG)
- TA: Arihant Jain/Aditya Shankar
- Simulate intersections where each node/agent acts as a traffic controller. When traffic builds up, agents negotiate who gets the green signal using lightweight protocols (e.g., token-passing, priority-based arbitration).
- Agent Capabilities: Local vehicle sensing, rule-based negotiation

Project 11: Feedback based speed control of a DC motor

- Faculty mentor: Harikumar Kandath (RRC)
- TA: Chetanya
- To demonstrate the principle of feedback control. The setup contains a DC motor with input voltage controlled through arduino, RPM sensor/encoder attached to motor shaft. The output of the RPM sensor/encoder is given to arduino device. The objective is to control the rotational speed/position of the motor as per the desired reference. Different feedback control laws can be tested using this setup.

Project 12: Obstacle avoidance for ground robot

- Faculty mentor: Harikumar Kandath (RRC)
- TA: Chetanya
- To implement obstacle avoidance for a ground robot navigating in an environment.
- The detection of the obstacle is performed using a range sensor. The range measured from the sensor is used to control the servo motor that drives the wheel of the robot.

Project 13: Gas Analyzer for Industrial Safety

Faculty: Abhishek Shrivastava

TA: Inesh

The aim is to design a portable gas analyzer that can detect toxic gases in industrial environments, display the results locally, trigger alarms when unsafe levels are detected, and transmit data wirelessly using LoRa for remote monitoring.

Deliverables:

- Algorithm Enhancement – testing advanced methods (e.g., polynomial fitting, machine learning) to improve accuracy of gas concentration estimation.
- Field Trials – Conduct extensive testing in varied industrial environments to validate detection accuracy, response time, and communication reliability.
- Power Optimization – Improve power management for longer device operation, especially under continuous monitoring.
- System Integration – Build a centralized dashboard for real-time visualization of gas levels from multiple devices over LoRa.
- Refinement for Deployment – Miniaturize the design, ruggedize the casing for industrial use, and move toward certification/standard compliance.

Project 14: Audio-Guided Autonomous Microbot System

Faculty: Abhishek Shrivastava

TA: Inesh

The aim is to design an intelligent microbot system equipped with MEMS microphone arrays capable of detecting, localizing, and following specific audio signatures (such as hand claps, voice commands, or predefined sound patterns) in real time. The system shall include autonomous navigation capabilities, secure wireless communication for remote operation, and a comprehensive GUI-based control console for mission planning, real-time monitoring, and manual override functionality.

Deliverables:

- **Audio Processing & Localization:** Develop signal processing and machine learning algorithms using MEMS microphone arrays for sound source detection, classification, and triangulation with beamforming in noisy environments
- **Autonomous Navigation:** Implement path planning and sensor fusion for obstacle-avoiding navigation toward sound sources using audio and proximity sensor data.
- **Secure Communication:** Enable encrypted LoRa/BLE communication with authentication and integrity checks for secure control and data transmission.
- **GUI Control Console:** Build a user interface for real-time status monitoring, sensor visualization, mission setup, actuator tuning, and emergency control.

QIDK-based projects

Project 1: GVP monitoring

- Faculty Guide: Sachin Chaudhari
- TA: Rahul Kumar
- Garbage vulnerable points (GVPs) are locations where waste is dumped without proper disposal. These sites produce foul odors serve as breeding grounds for pests and stray animals, cause soil and water pollution, and create unsanitary conditions in localities.
- Create a QIDK-based GVP monitoring solution using computer vision and IoT technologies in urban areas.
- You can determine different parameters such as amount of garbage, collection schedules, dumping patterns and more....

Project 2: Construction Activity

- Faculty Guide: Saikiran Bulusu
- TA: Srujan
- At construction sites, insufficient inspection and supervision often result in decreased worker efficiency and delays in project completion. Supervisors struggle to monitor each worker individually, particularly across large or complex sites. This lack of consistent oversight reduces accountability and makes it difficult to maintain productivity levels throughout the workforce.
- To address this issue, develop a QIDK edge computing-based Construction Activity Monitoring Solution using computer vision and image processing algorithms.
 - Implement worker detection and location tracking on-site.
 - Identify daily progress variations within the construction zone and correlate them with the activity levels of different worker groups.

Project 3: Sustainable Agentic AI in the Edge

- Faculty: Karthik Vaidhyanathan
- TA: Aditya Kishore
- Agentic AI is showing promise where a group of agents powered by AI models can collaborate and perform various tasks. Among these, one thing that is getting more commonly used are LLM agents where agents can leverage LLMs and different tools to achieve their intended task.
- Develop an agentic AI storyteller application using QIDK as a use case to demonstrate the possibility of engineering agentic AI at the edge using small language models and lightweight vision models
- The goal will be to benchmark the performance metrics of the agentic ai system in the edge – Mainly CPU utilization and accuracy

Project 4: AI Sustainability Assessment Framework at the Edge

- Faculty: Karthik Vaidhyanathan
- Support Person: Ms. Akhila Matathammal
- TA: Aditya Kishore

- LLMs have a big impact on the global carbon footprint. To this, SLMs have been shown to be a good alternative. However, there is a lack of framework for assessing sustainability impact at design time.

- In this work, the goal will be to benchmark different SLMs for multiple tasks: i) Question answer; ii) Retrieval augmented Generation and iii) summarization on parameters like energy, accuracy, latency
 - As a next step, build a simple tool that given a task and Quality of service expectations, can suggest a model to be deployed.

Project 5: Sustainable Multimodal AI at the Edge

- Faculty: Priyesh Shukla
- TA: Udith
- Implement/deploy Bit-net (1-bit LLM & extend to VLM) and multi-modal SLAM at QIDK platform with quantization-aware training and inference for energy efficiency
- Use case for above implementation: Multimodal SLAM (for insect-sized drones) with visual and acoustic data as well as visual and text data
 - Data collection (through microphone, monocular cam and texts)
 - Lightweight model development and optimization
 - Model deployment at QIDK

Project 6: Pose Detection

- Faculty: Aftab Hussain
- TA: Rahul
- Use video to determine the sitting posture of a person on a chair – sitting straight, slouching, cross-legged etc.
 - If done, provide suggestions if not sitting properly.

Project 7: Voice controlled Smart Home Assistant

- Faculty mentor: Suresh Purini
- TA: Aditya Shankar
- Develop a voice controlled smart home assistant which can
 - Interprets voice commands and converts to function calls with appropriate parameters
 - Switch on/off the lights in bedrooms
 - Use QIDK platform

Projects 8: AI powered Tutor

- Guide: Ravi (Qualcomm)
 - Faculty: Karthik Vaidhyanathan
 - TA: Aditya Shankar
-
- Description: Build a tutoring app that answers questions, generates quizzes, and explains concepts using generative text models.

Projects 9: On-Device Image Inpainting and Editing Tool

- Guide: Ravi (Qualcomm)
 - Faculty: Deepak
 - TA: Udith
-
- Description: Create an app that allows users to remove objects from images and fill gaps using generative models.

Projects 10: Gesture-Controlled Smart Interface

- Guide: Ravi (Qualcomm)
- Faculty: Aftab
- TA: Srujan

- Description: Build a gesture recognition system that triggers generative responses based on hand movements.

Course Logistics

Project Distribution

- Fill the form and preferences by 05 Aug. 2025
 - <https://forms.office.com/r/iBPqfXfEhB>
- Lottery followed by preferences
- Projects allotted by 07 Aug. 2025 by Head TAs

Course Portal

- Moodle
 - Notifications
 - News
 - Discussion Forum
 - Submissions

Evaluation Plan

- Mark distributions: 30-70
- 30: 10 (midsem) + 20 (final)
- 70 for labs (tentative):
 - Evaluation for labs 1-3: 10 (by respective faculty/mentors)
 - Evaluation for labs 4-6: 15 (by Sachin+Aftab+Deepak)
 - Evaluations for labs 7-9: 15 (by respective faculty)
 - Overall: 30 (Externals + Deepak+Aftab+Sachin+Internal Faculty)
- Grading: Relative (TBD)

Attendance Policy

- Labs
 - Once per week on Monday 2-5 pm
 - Mandatory
- Classes
 - Once per week on Thursday 2-3.30 pm
 - Not mandatory

Reimbursement for Project Sensors

- Most of the components are expected to be in the lab
- Students will be reimbursed for the project-specific sensors and other electronic components which needs to be additionally procured
 - Please consult your faculty and TA before purchasing the project-specific sensors
 - Get a written approval from respective faculty over email cc:ing Lab Manager (Mr. Prashant Ranjan) and TA
- Maximum up to Rs. 5 K per team!
 - In case going above it, need to request the faculty coordinators with requested budget with justification.
 - You can always buy sensors on your own and add to the project
- Bills should be proper GST bills (not quotations/estimates) and in the name of IIIT Hyderabad.
- Note that all the reimbursement bills must be submitted all components should be returned by 8 Dec. 2025.

Questions?

- That's all for today!